

A norm-convergence partial answer for Bourgain–uo Cauchy sequences in ℓ_p

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Status

This packet records a candidate partial result for Question 5.2 of Kania and Swaczyna’s paper *Bourgain–uo sequential completeness in vector lattices* [1]. The source question asks whether Buo-Cauchy sequences converge in ℓ_p , $1 < p < \infty$. We prove that they always converge in the norm of ℓ_p , for every $1 \leq p < \infty$. This does not by itself prove order convergence or Buo-convergence, so the packet is classified as a partial result unless the intended reading of the question is only norm convergence.

Source Question

The open question appears on page 12 of the source PDF.

5. OPEN PROBLEMS

Question 5.1. Characterise those Banach lattices for which every norm bounded Buo-Cauchy sequence is uo-convergent. How does this property relate to the bounded uo-completeness studied in [9]?

Question 5.2. Do Buo-Cauchy sequences converge in

- ℓ_p for $1 < p < \infty$?
- the Morrey spaces $\mathcal{M}^{p,\kappa}(\mathbb{T})$ and in Besov/Triebel–Lizorkin spaces $B_{p,q}^s(\mathbb{T}), F_{p,q}^s(\mathbb{T})$ in the usual parameter ranges?

The relevant bullet asks:

Do Buo-Cauchy sequences converge in ℓ_p for $1 < p < \infty$?

Definitions

We use the source paper’s definition: a sequence (x_n) in a vector lattice E is Buo-Cauchy if, for every strictly increasing sequence (n_k) , the consecutive differences

$$x_{n_{k+1}} - x_{n_k}$$

converge to 0 in order in E .

In ℓ_p , $1 \leq p < \infty$, order convergence to 0 implies norm convergence to 0. This is the standard order continuity of the ℓ_p -norm.

Proof Intuition

The Buo-Cauchy condition is stronger than merely saying that the original successive differences are small: it tests the successive differences of every subsequence. In any metric space, a sequence is Cauchy exactly when no subsequence can be arranged to have infinitely many large consecutive jumps. In ℓ_p , order-null difference sequences are norm-null, so the Buo-Cauchy subsequence test rules out those large metric jumps.

Main Result

Theorem 1. *Let $1 \leq p < \infty$. Every Buo-Cauchy sequence in ℓ_p is norm Cauchy. Consequently, every Buo-Cauchy sequence in ℓ_p converges in the norm topology of ℓ_p .*

Proof. Let (x_n) be a Buo-Cauchy sequence in ℓ_p . Suppose, toward a contradiction, that (x_n) is not Cauchy in norm. Then there is $\varepsilon > 0$ such that for every N there are $m, n \geq N$ with

$$\|x_n - x_m\|_p \geq \varepsilon.$$

By induction we may choose strictly increasing indices

$$r_1 < r_2 < r_3 < \cdots$$

such that

$$\|x_{r_{2j}} - x_{r_{2j-1}}\|_p \geq \varepsilon \quad (j \in \mathbb{N}).$$

Indeed, after r_{2j-2} has been chosen, apply non-Cauchyness with $N > r_{2j-2}$ and relabel the resulting pair in increasing order.

Apply the Buo-Cauchy hypothesis to the subsequence (x_{r_k}) . By definition, the sequence of consecutive differences

$$d_k := x_{r_{k+1}} - x_{r_k}$$

converges to 0 in order in ℓ_p . Since the ℓ_p -norm is order continuous, $d_k \rightarrow 0$ in norm. In particular,

$$\|d_{2j-1}\|_p = \|x_{r_{2j}} - x_{r_{2j-1}}\|_p \rightarrow 0,$$

contradicting the lower bound $\|x_{r_{2j}} - x_{r_{2j-1}}\|_p \geq \varepsilon$. Thus (x_n) is norm Cauchy. Since ℓ_p is norm complete, (x_n) converges in norm to some element of ℓ_p . $\square$

Verification Notes

The proof uses only two standard facts:

1. In a metric space, non-Cauchyness produces a subsequence with infinitely many consecutive jumps bounded below by a fixed $\varepsilon > 0$.
2. In ℓ_p , $1 \leq p < \infty$, order convergence implies norm convergence, because the norm is order continuous.

The source paper already proves a different partial fact: if a Buo-Cauchy sequence in ℓ_p has coordinatewise limit x , then $x \in \ell_p$ [1, Lemma 3.5]. The theorem above is independent of that lemma, and gives norm convergence directly.

Limitations and Novelty Check

This packet does not prove that the norm limit is also the order limit. Norm convergence in ℓ_p is strictly weaker than order convergence, so the order/Buo-convergence version of Question 5.2 remains open here.

Cheap run indexes contained no prior packet or attempt for arXiv:2512.14949. A bounded web search for exact phrases from the question and the source title found the source paper and adjacent Bourgain–uo/free-lattice material, but no obvious later resolution of the ℓ_p bullet. Because the result is a partial norm-convergence observation, novelty confidence is moderate rather than high.

Human Review Recommendation

Verify the intended topology in Question 5.2. If “converge in ℓ_p ” means norm convergence, the ℓ_p bullet is answered affirmatively by the theorem above. If it means order convergence or Buo-convergence, this is a useful partial result and the remaining domination/order-boundedness step should be reviewed separately.

References

- [1] T. Kania and J. Swaczyna, *Bourgain–uo sequential completeness in vector lattices*, arXiv:2512.14949, 2025.